Quantification: 701_z26c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	31
interior lacuna count	20
frame-edge lacuna count	11

Interior lacunae only: mean, median and sample SD (n = 20)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2447.35	1515.5	2769.53	px^2
major_axis_length_px	lacuna shape	83.2547	67.942	36.8475	px
minor_axis_length_px	lacuna shape	40.3949	33.3902	22.1828	px
aspect_ratio	lacuna shape	2.2409	2.1031	0.7643	unitless
eccentricity	lacuna shape	0.8273	0.8798	0.1679	unitless
solidity	lacuna shape	0.7291	0.7314	0.0941	unitless
roots_count	headline	9.9	8.5	4.2538	count
ring_length_r30_px	headline	402	403.5	157.567	px
ring_length_r60_px	reported	796.15	768	369.784	px
owned_length_px	ownership-dependent	1182.73	869.482	1052.36	px
edge_count	ownership-dependent	80.3	59.5	70.6378	count
mean_edge_length_px	ownership-dependent	14.5653	14.7155	1.9932	px
perimeter_px	normalised	323.693	275.484	176.847	px
ring_area_r30_px2	normalised	8329.55	7452	3086.68	px^2
ring_area_r60_px2	normalised	16077	15961	6256.08	px^2
in_frame_fraction_r30	normalised	0.9973	1	0.0121	unitless
in_frame_fraction_r60	normalised	0.9907	1	0.0328	unitless
ring_density_r30	normalised	0.0487365	0.045382	0.012904	px^-1
ring_density_r60	normalised	0.0490185	0.0457414	0.0135523	px^-1
roots_per_100px_perimeter	normalised	3.3746	3.4352	0.9623	per 100 px
ring_attached_length_r30_px	network v2	348.15	363.5	150.948	px
ring_attached_length_r60_px	network v2	594.8	572	357.586	px
ring_length_w_r30_px	network v2	453.14	453.534	177.452	px
ring_length_w_r60_px	network v2	895.704	857.497	417.344	px
sholl_crossings_r10	network v2	9.95	9.5	4.4186	count
sholl_crossings_r20	network v2	10.5	8.5	4.383	count
sholl_crossings_r30	network v2	10.3	10	5.2425	count
loop_count	loops, report only	5.05	1	8.9706	count
loop_edge_share	loops, report only	0.1549	0.1218	0.166	share
ring_width_mean_r30_px	width	5.9605	5.9174	1.3216	px

Per field
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measure	value	units
analysed_area_px2	900357	px^2
total_skeleton_length_px	43213	px
canalicular_length_density_per_px	0.0479954	px^-1
junction_count	1253	count
junction_density_per_px2	0.00139167	px^-2
lacuna_count	31	count

lacunae_per_px2	3.443e-05	px^-2
skeleton_component_count	682	count
mean_component_length_px	63.3622	px
median_component_length_px	14	px
field_length_density_w_per_px	0.0539535	px^-1
field_density_without_flagged_per_px	0.0481251	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	286	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0479954	px^-1
field_length_density_with_unattached_per_px	0.0483131	px^-1
loop_count	163	count
loop_edge_share	0.2178	share
edge_count	3168	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.4376	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	28572	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6622	126.492	79.7855	1.5854	0.776	0.7588	0.7952	43.4012	272.285	True
2	24115	285.31	118.544	2.4068	0.9096	0.7591	1.2945	92.9897	132.363	True
3	8121	199.992	79.1327	2.5273	0.9184	0.6393	0.943	189.47	54.1918	True
4	6792	139.701	72.9543	1.9149	0.8528	0.7645	1.2651	31.6926	963.876	True
5	2287	125.157	32.9459	3.7989	0.9647	0.6065	0.5498	75.4101	463.222	False
6	3999	122.318	44.0398	2.7774	0.9329	0.877	0.1379	179.418	1003.78	True
7	1531	85.4004	40.5987	2.1035	0.8798	0.5916	1.0809	183.008	698.054	False
8	2493	118.127	38.0041	3.1083	0.9468	0.6054	-1.3995	230.773	666.071	False
9	5110	133.797	67.4988	1.9822	0.8634	0.6757	1.2061	267.96	890.703	False
10	1782	62.6142	47.9495	1.3058	0.6431	0.682	1.1297	243.333	206.608	False
11	1422	68.8124	28.5123	2.4134	0.9101	0.8335	1.1259	314.225	198.443	False
12	1652	67.0082	31.8663	2.1028	0.8797	0.8959	0.9049	359.245	256.938	False
13	2608	67.6663	54.9533	1.2313	0.5835	0.8644	0.6706	338.284	24.3976	True
14	30788	334.972	147.951	2.2641	0.8972	0.683	0.7858	446.787	150.129	True
15	12693	158.381	111.247	1.4237	0.7118	0.8419	-1.4075	620.143	187.231	False
16	967	67.0717	21.9089	3.0614	0.9451	0.7662	-1.3191	319.159	644.63	False
17	8889	153.266	79.6369	1.9246	0.8544	0.9019	1.3702	455.699	960.443	True
18	821	63.8935	20.0569	3.1856	0.9495	0.7609	0.1039	594.211	741.577	False
19	4845	153.224	58.0129	2.6412	0.9256	0.6393	-0.5	676.391	794.742	False
20	2272	79.8087	42.6154	1.8728	0.8455	0.7881	-1.0855	686.797	269.451	False
21	640	42.0108	20.3754	2.0618	0.8745	0.8625	1.0676	716.059	885.861	False
22	656	33.1626	30.2759	1.0953	0.4081	0.801	1.2644	753.977	863.463	False
23	883	72.1056	21.4511	3.3614	0.9547	0.6761	-1.2791	751.882	124.115	False
24	1013	51.2657	31.9563	1.6042	0.7819	0.7251	-0.0492	792.697	1003.49	True

25	1435	61.7195	33.8344	1.8242	0.8363	0.8112	-1.4959	787.26	645.571	False
26	4376	111.203	63.8724	1.741	0.8186	0.6554	-1.0966	845.546	482.61	False
27	1500	55.4417	51.2017	1.0828	0.3835	0.6268	-1.2338	890.901	936.497	False
28	718	51.4892	21.7937	2.3626	0.906	0.7349	-1.184	917.613	557.102	False
29	864	54.6671	23.8765	2.2896	0.8996	0.7279	-1.1377	941.147	439.961	False
30	3941	116.706	45.969	2.5388	0.9192	0.89	-1.4794	1004.56	375.487	True
31	2384	124.611	32.5415	3.8293	0.9653	0.6932	-1.5289	1010.89	775.227	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	344	618	349.421	24	14.5592	5.9536
2	26	791	905	814.482	66	12.3406	4.6531
3	11	387	453	328.735	22	14.9425	6.2389
4	14	492	731	432.676	39	11.0943	5.4987
5	19	555	1192	748.027	57	13.1233	6.735
6	11	382	548	859.11	51	16.8453	6.1919
7	11	360	613	403.706	29	13.9209	5.9174
8	12	464	847	1124.78	72	15.6219	5.0033
9	18	567	1202	2074.78	142	14.6111	6.7551
10	9	309	386	266.652	18	14.814	5.9108
11	7	191	196	213.208	14	15.2292	3.8149
12	4	202	436	98.1543	5	19.6309	4.3661
13	1	27	84	7.8284	1	7.8284	9.1513
14	11	532	999	333.22	24	13.8842	4.3419
15	12	382	711	794.014	54	14.704	5.4258
16	9	294	549	412.191	32	12.881	6.4769
17	7	469	1252	1648.74	109	15.126	5.4301
18	6	297	639	944.95	62	15.2411	7.1772
19	10	624	1118	2187.42	130	16.8263	3.8627
20	8	333	825	1203.98	77	15.6361	6.7465
21	5	164	383	151.569	17	8.9158	4.1534
22	7	190	292	277.723	22	12.6238	5.4386
23	8	486	1127	4152.97	282	14.7269	7.8175
24	6	190	339	235.48	12	19.6234	5.8946
25	8	484	1080	2029.5	141	14.3936	7.4781
26	18	741	1579	2512.16	178	14.1132	5.823
27	12	475	1007	1101.99	74	14.8918	5.9812
28	7	497	1096	2287.63	154	14.8547	8.3653
29	8	425	645	669.132	46	14.5463	
30	15	387	750	1240.8	82	15.1317	4.193
31	9	276	681	96.3553	7	13.765	

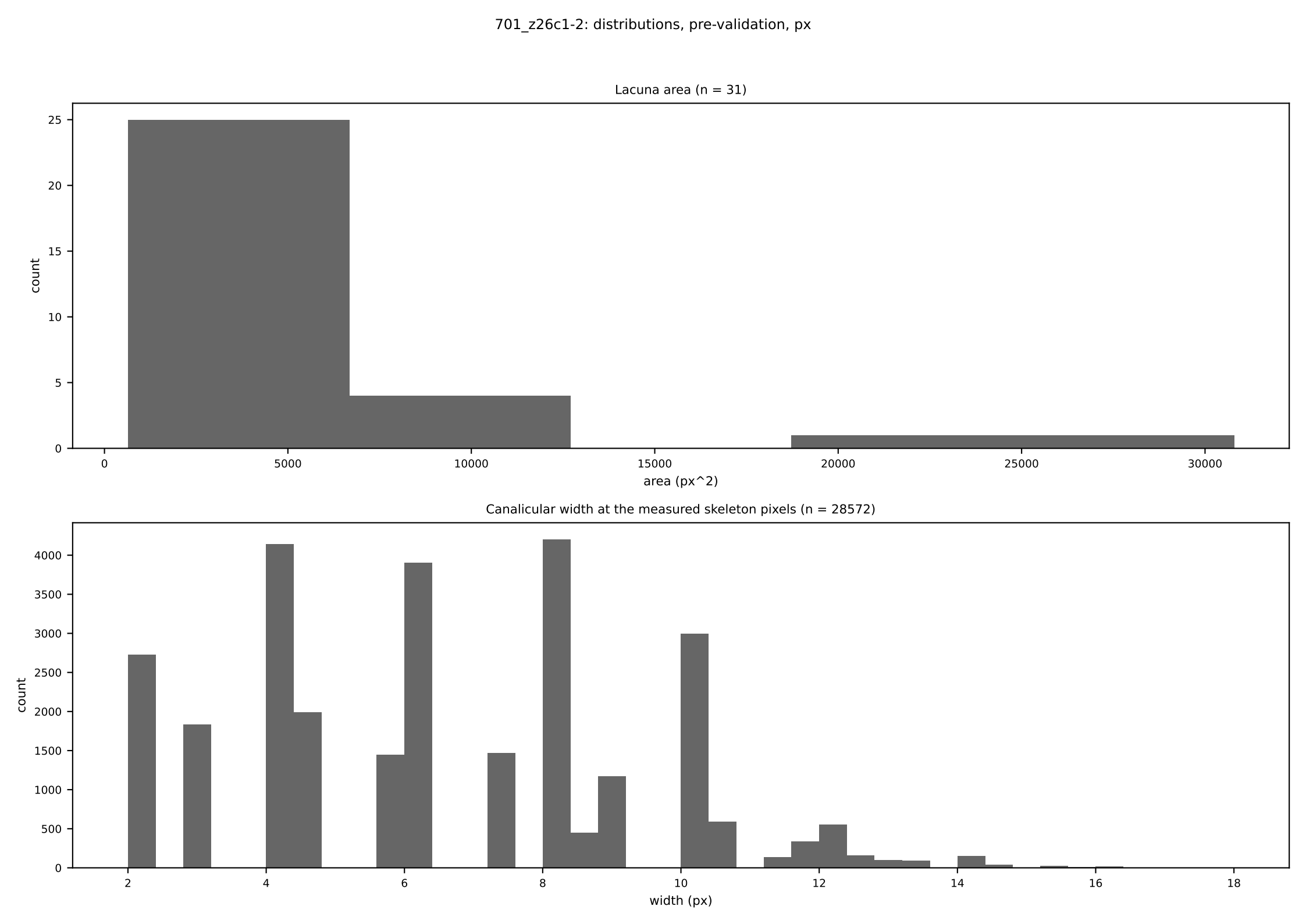
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	751.134	8143	12926	0.7412	0.6831	0.0422449	0.0478106	1.997
2	1296.59	17823	20380	0.8324	0.6918	0.0443809	0.0444063	2.0053
3	652.802	10422	12277	0.7803	0.7374	0.037133	0.0368983	1.685
4	561.257	7623	12937	0.5264	0.4563	0.0645415	0.0565046	2.4944
5	459.683	11984	25145	0.9457	0.8575	0.0463117	0.047405	4.1333
6	376.019	7018	10546	0.6142	0.5806	0.0544315	0.0519628	2.9254
7	322.475	7086	14762	1	1	0.0508044	0.0415255	3.4111
8	439.713	9174	16100	1	1	0.0505777	0.0526087	2.7291
9	541.524	14452	28872	1	1	0.0392333	0.041632	3.324
10	345.031	6744	9972	1	1	0.0458185	0.0387084	2.6085
11	202.35	4921	5276	1	1	0.0388133	0.0371494	3.4593

12	196.522	5790	10562	1	1	0.0348877	0.0412801	2.0354
13	273.48	2279	3229	0.6705	0.6272	0.0118473	0.0260142	0.3657
14	1396.63	22273	32588	0.9271	0.8877	0.0238854	0.0306555	0.7876
15	659.572	11568	19301	1	1	0.0330221	0.0368375	1.8194
16	188.244	7262	16365	1	1	0.0404847	0.0335472	4.781
17	546.701	11662	25518	0.7611	0.7175	0.0402161	0.0490634	1.2804
18	163.723	6608	14115	1	1	0.0449455	0.045271	3.6647
19	660.199	14216	24193	1	1	0.0438942	0.0462117	1.5147
20	273.019	7642	15822	1	1	0.043575	0.0521426	2.9302
21	128.202	4822	9334	1	1	0.0340108	0.0410328	3.9001
22	133.338	4666	9184	1	1	0.0407201	0.0317944	5.2498
23	277.948	7921	17655	1	1	0.0613559	0.0638346	2.8782
24	208.386	5088	9797	0.7064	0.6319	0.0373428	0.0346024	2.8793
25	215.7	7866	19003	1	1	0.0615306	0.0568331	3.7089
26	605.222	12635	22738	1	1	0.0586466	0.0694432	2.9741
27	307.298	9361	20597	1	0.9574	0.0507424	0.0488906	3.905
28	167.823	6421	14132	1	1	0.0774023	0.0775545	4.1711
29	186.279	5452	8413	1	1	0.077953	0.0766671	4.2946
30	332.007	6043	11810	0.5904	0.5692	0.064041	0.0635055	4.518
31	413.233	6749	15846	0.5979	0.5709	0.0408949	0.0429761	2.178

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	304	368	379.563	684.215	8	5	8
2	697	772	886.838	1019.55	18	25	5
3	349	355	439.262	511.718	13	14	1
4	434	457	561.76	831.21	14	9	12
5	494	692	631.843	1347.07	20	10	12
6	316	488	430.09	617.357	10	15	7
7	283	328	409.534	689.428	9	8	9
8	407	569	519.032	946.11	13	12	8
9	533	915	647.127	1357.24	16	18	17
10	209	234	338.279	420.978	7	7	3
11	145	145	216.894	223.551	5	4	3
12	135	159	231.108	496.002	6	8	5
13	8	8	30.799	97.9117	1	0	3
14	327	335	597.571	1118.06	12	14	13
15	315	453	429.505	793.997	10	8	9
16	218	321	326.936	612.115	6	7	7
17	379	801	519.475	1402.79	14	16	13
18	241	492	347.363	731.997	6	8	12
19	466	856	698.813	1276.07	15	17	10
20	320	575	374.492	920.997	8	9	11
21	126	126	181.297	429.191	4	7	6
22	190	240	211.095	317.622	5	7	2
23	461	1057	552.404	1284.22	9	14	14
24	146	192	210.167	381.333	5	3	6
25	448	941	540.99	1205.02	11	15	18
26	654	1416	827.299	1756.07	17	18	20
27	407	726	538.103	1134.25	11	13	18
28	490	1015	563.132	1246.66	11	15	12
29	421	636	477.563	725.487	10	5	10
30	360	669	422.664	839.784	10	10	9
31	107	107	303.007	755.269	5	5	13


701_z26c1-2: distributions, pre-validation, px
Lacuna area (n = 31)
count
area (px^2)
0
5
10
15
20
25
0
5000
10000
15000
20000
25000
30000
Canalicular width at the measured skeleton pixels (n = 28572)
count
width (px)
0
500
1000
1500
2000
2500
3000
3500
4000
2
4
6
8
10
12
14
16
18